FATIMA JINNAH WOMEN UNIVERSITY, RAWALPINDI

DEPARTMENT OF COMPUTER SCIENCE

LAB 02

Hands-On Git and Version Control

Submitted by

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DevOps (CS-649)

BCS VII

Github Username

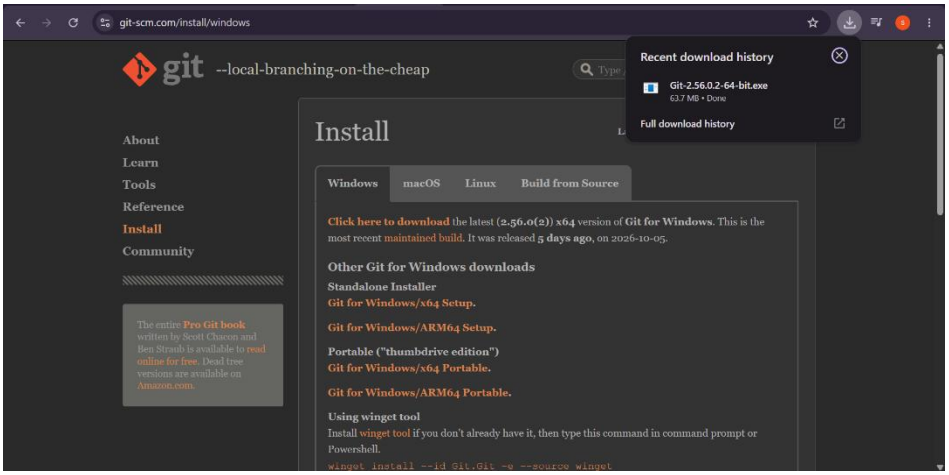
SyedaAlishba12

Introduction

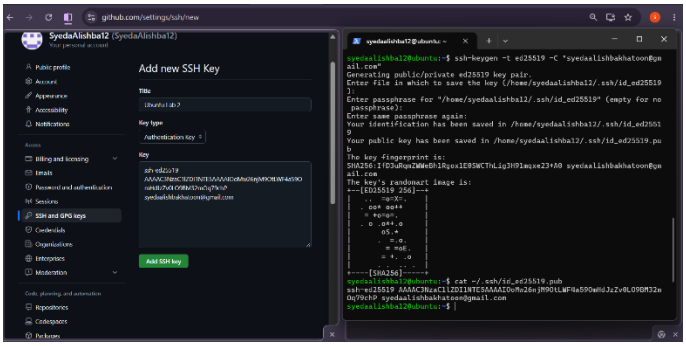
In this lab, I learned how to use Git and GitHub for version control. I practiced creating repositories, making commits, working with branches, pushing changes to GitHub, and merging branches. I also learned about pull requests and how they are used to review code before merging.

1. Git Installation and Setup

First, I checked Git installation and configured my GitHub username and email. I generated an SSH key and added the public key to my GitHub account. After that, I used SSH to connect to GitHub and clone my practice repository.




```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.name "Syeda Alishba Khatoon"
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.email "syedaalishbakhatoon@gmail.com"
```



2. Repository Setup

I practiced checking the Git repository folder and understanding the .git directory. I removed the existing Git configuration, initialized Git again, and created my first commit. Then I connected the local repository to GitHub and pushed the changes.

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git status
fatal: not a git repository (or any of the parent directories): .git
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/syedaalishba12/DevOps_Lab2_Practice/.git/
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ echo "# Lab02 Git Practice" >> README.md
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial commit"
[master (root-commit) c5154aa] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:syedaalishba12/DevOps_Lab2_Practice.git
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git branch -M main
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 246 bytes | 35.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

3. Working with Files and Commits

I created and edited files such as README.md, notes.txt, and main.py. I used Git commands to check the status of my files and save changes through commits.

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some test notes" >> notes.txt
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        notes.txt

nothing added to commit but untracked files present (use "git add" to track)
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add notes.txt"
[main e5aad83] Add notes.txt
 1 file changed, 1 insertion(+)
 create mode 100644 notes.txt
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

4. Branching and Merging

I created different branches to practice working on separate tasks. I created feature and bug-fix branches, made changes, and merged them into the required branches. I also practiced pushing branches to GitHub and deleting branches after merging.

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git merge feature/login
Updating 770896c..08267c4
Fast-forward
 login_feature.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login_feature.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
   770896c..08267c4  develop -> develop
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git checkout bugfix/login-error
Switched to branch 'bugfix/login-error'
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ echo "# Login error fix practice" > login_error_fix.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch bugfix/login-error
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        login_error_fix.py

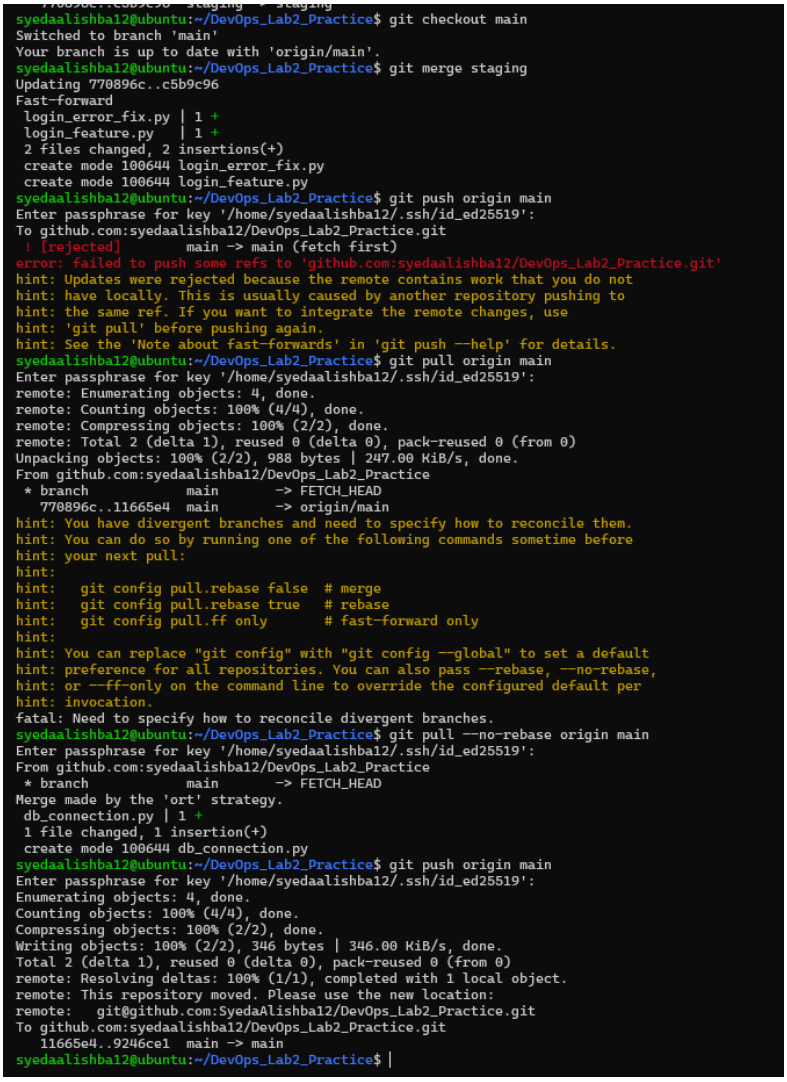
nothing added to commit but untracked files present (use "git add" to track)
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git add login_error_fix.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add login error fix practice file"
[bugfix/login-error 62d5799] Add login error fix practice file
 1 file changed, 1 insertion(+)
 create mode 100644 login_error_fix.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin bugfix/login-error
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 404 bytes | 134.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
   770896c..62d5799  bugfix/login-error -> bugfix/login-error
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git checkout develop
Switched to branch 'develop'
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git merge bugfix/login-error
Merge made by the 'ort' strategy.
 login_error_fix.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login_error_fix.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 329 bytes | 21.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
   08267c4..c5b9c96  develop -> develop
```

```
Switched to a new branch 'bugfix/login-error'
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/SyedaAlishba12/DevOps_Lab2_Practice/pull/new/develop
remote:
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 * [new branch]      develop -> develop
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin staging
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'staging' on GitHub by visiting:
remote:      https://github.com/SyedaAlishba12/DevOps_Lab2_Practice/pull/new/staging
remote:
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 * [new branch]      staging -> staging
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/login
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'feature/login' on GitHub by visiting:
remote:      https://github.com/SyedaAlishba12/DevOps_Lab2_Practice/pull/new/feature/login
remote:
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 * [new branch]      feature/login -> feature/login
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin bugfix/login-error
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'bugfix/login-error' on GitHub by visiting:
remote:      https://github.com/SyedaAlishba12/DevOps_Lab2_Practice/pull/new/bugfix/login-error
remote:
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 * [new branch]      bugfix/login-error -> bugfix/login-error
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
From github.com:syedaalishba12/DevOps_Lab2_Practice
 * branch            bugfix/user-auth-error -> FETCH_HEAD
 * [new branch]      bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add new function to main.py"
[feature-1 770896c] Add new function to main.py
 1 file changed, 4 insertions(+)
 create mode 100644 main.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

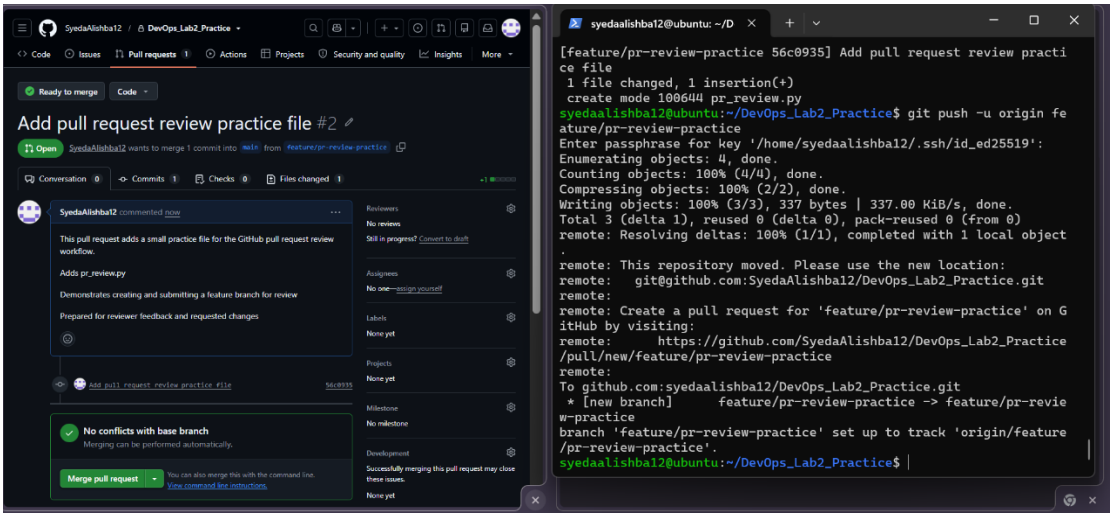
```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/db-connection
Switched to a new branch 'feature/db-connection'
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/db-connection
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 631 bytes | 126.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/SyedaAlishba12/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```




```
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating e2b9896..770896c
Fast-forward
 main.py | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 main.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```

5. Pull Request Practice

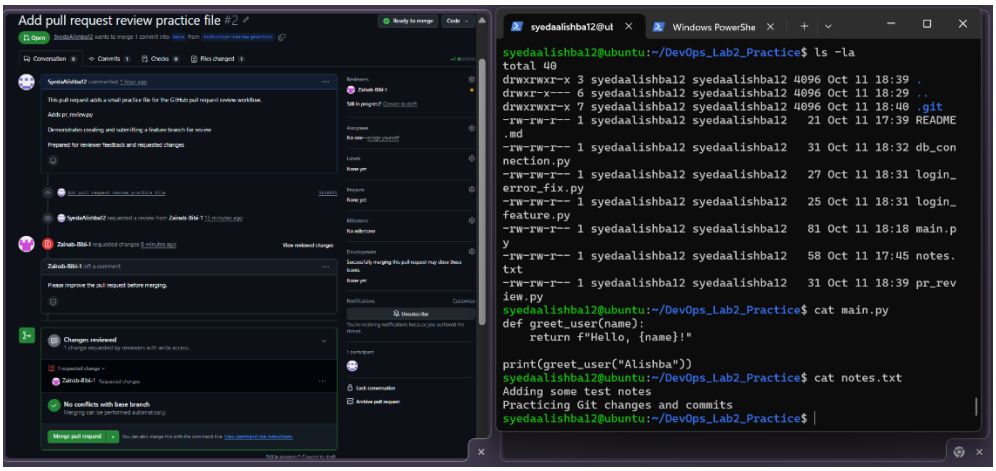
I created a pull request from my feature branch to the main branch. I added a title and description explaining the changes. This helped me understand how pull requests are used to submit changes for review before merging them into the main branch.

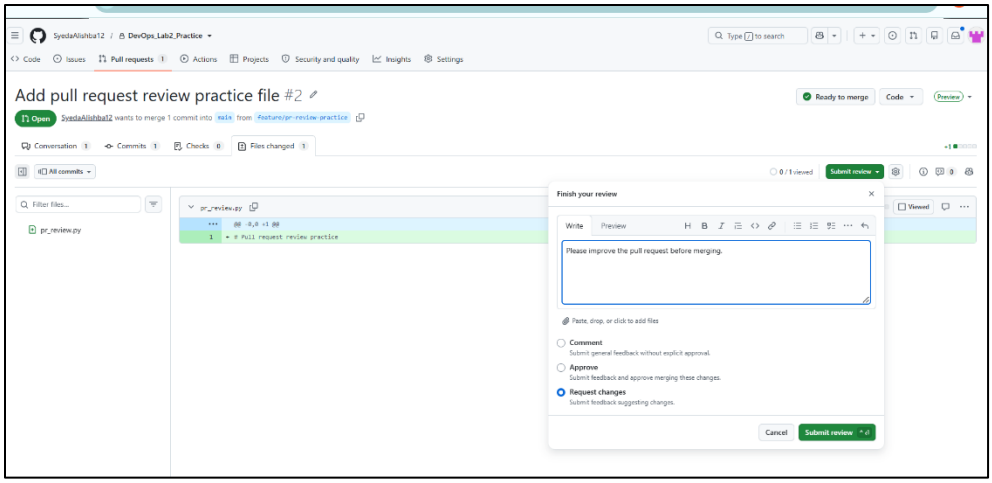


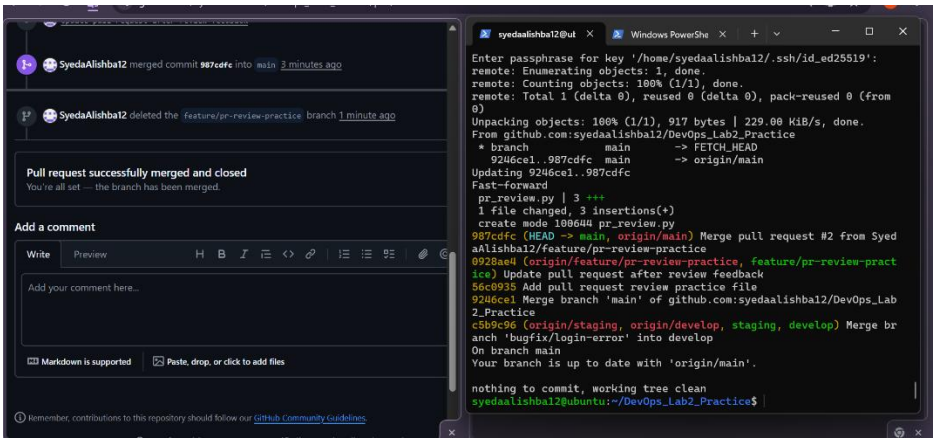
6. Pull Request Review and Feedback — Task 12

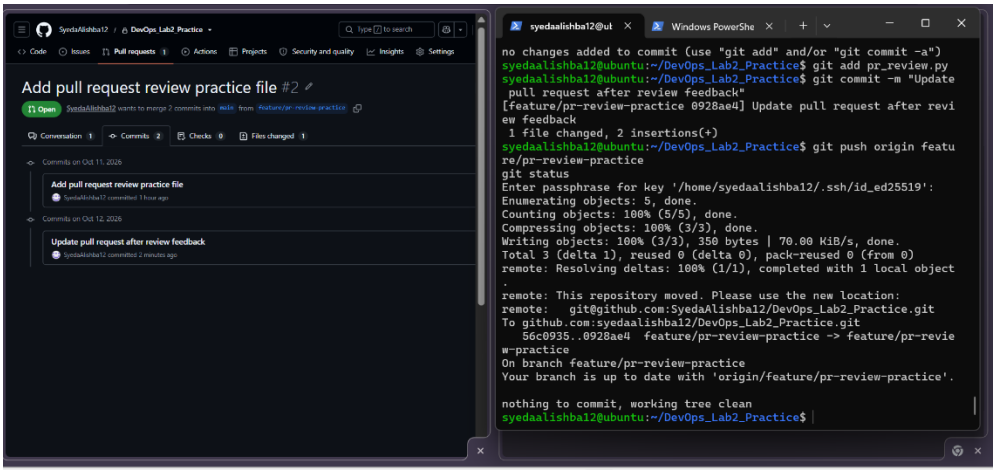
For this task, I practiced the complete pull request review process. I assigned a collaborator as the reviewer and requested feedback on my code. The reviewer submitted a request for changes, demonstrating how code can be reviewed before it is merged.

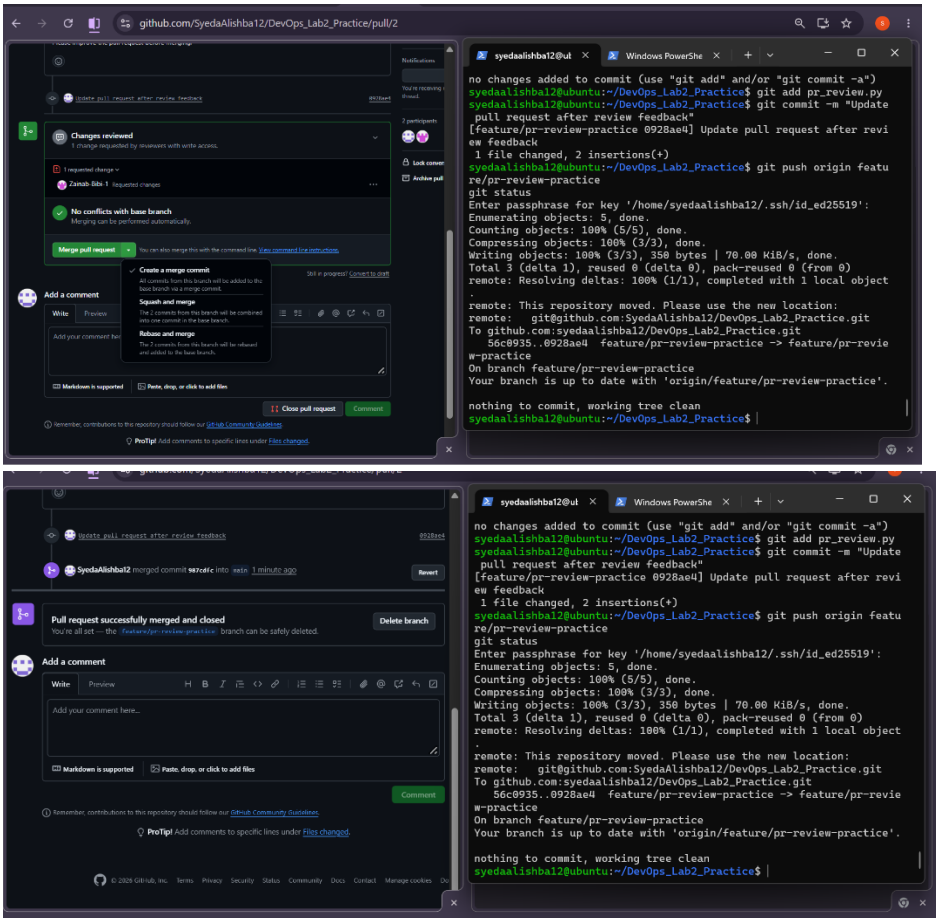
After receiving feedback, I updated pr_review.py and committed the changes. I then pushed the updated code to GitHub so the reviewer could see the latest version. Finally, I merged the pull request and deleted the feature branch.




Add pull request review practice file #2
Finish your review
Please improve the pull request before merging.
Comment
Approve
Request changes
Cancel
Submit review


SyedaAlishba12 merged commit 987cdfc into main 3 minutes ago
SyedaAlishba12 deleted the feature/pr-review-practice branch 1 minute ago
Pull request successfully merged and closed
Add a comment
Write
Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 917 bytes | 229.00 KiB/s, done.
From github.com:syedaalishba12/DevOps_Lab2_Practice
* branch main -> FETCH_HEAD
9246ce1..987cdfc main -> origin/main
Updating 9246ce1..987cdfc
Fast-forward
pr_review.py | 3 +++
1 file changed, 3 insertions(+)
create mode 100644 pr_review.py
987cdfc (HEAD -> main, origin/main) Merge pull request #2 from SyedaAlishba12/feature/pr-review-practice
0928ae4 (origin/feature/pr-review-practice, feature/pr-review-practice) Update pull request after review feedback
56c0935 Add pull request review practice file
9246ce1 Merge branch 'main' of github.com:syedaalishba12/DevOps_Lab2_Practice
c5b9c96 (origin/staging, origin/develop, staging, develop) Merge branch 'bugfix/login-error' into develop
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$


Add pull request review practice file #2
Add pull request review practice file
Update pull request after review feedback
no changes added to commit (use "git add" and/or "git commit -a")
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git add pr_review.py
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Update pull request after review feedback"
[feature/pr-review-practice 0928ae4] Update pull request after review feedback
1 file changed, 2 insertions(+)
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature/pr-review-practice
git status
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 350 bytes | 70.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote: git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
56c0935..0928ae4 feature/pr-review-practice -> feature/pr-review-practice
On branch feature/pr-review-practice
Your branch is up to date with 'origin/feature/pr-review-practice'.
nothing to commit, working tree clean
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$

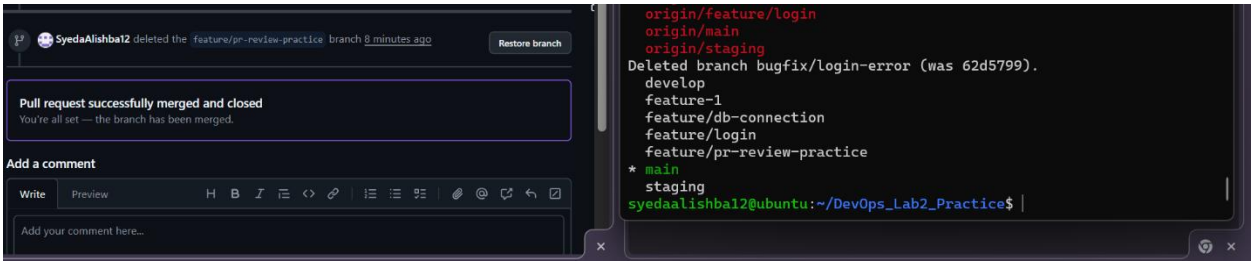


7. Remote Branch Deletion — Task 13

In this task, I practiced deleting remote branches through both the GitHub website and the terminal. I verified the remote branch list after deletion and removed the corresponding local branch after confirming that its changes had been merged into the main branch.

This task helped me understand the difference between deleting a branch on GitHub and deleting a local branch in my repository.

```
  staging
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete bugfix/login-error
Enter passphrase for key '/home/syedaalishba12/.ssh/id_ed25519':
remote: This repository moved. Please use the new location:
remote:   git@github.com:SyedaAlishba12/DevOps_Lab2_Practice.git
To github.com:syedaalishba12/DevOps_Lab2_Practice.git
 - [deleted]         bugfix/login-error
syedaalishba12@ubuntu:~/DevOps_Lab2_Practice$
```



8. Files Included in the Submission

My submission folder contains the lab README, Python files, notes, screenshots, and this solution report.

- README.md
- main.py
- notes.txt
- screenshots/
- Lab2_Solution.pdf

Conclusion

This lab helped me understand the Git and GitHub workflow, including repository setup, commits, branching, merging, and remote repository management. I also practiced creating pull requests, assigning reviewers, responding to requested changes, updating code, merging reviewed changes, and deleting branches. These activities helped me understand how version control supports collaboration and organized software development.